The Problem: Deadly Weld Defects

Weld defects can lead to catastrophic failures in critical structures. Current manual visual inspections are slow, subjective, and prone to errors.

> "Weld defects kill. A single missed crack in a pipeline, a bridge joint, or a pressure vessel can lead to catastrophic failure."

Our challenge: automatically detect and classify weld defects in real-time using raw sensor data, audio, and video, with confidence scores.

Dataset: **1,551** labeled welding runs across 7 classes (1 good, 6 defect types) with real industrial multimodal data.

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Physics-Based Feature Engineering

Instead of a black-box deep learning model, we adopted a physics-first approach, asking: "What does a weld defect actually sound like and look like?"

Audio (180 features)

- **Sub-band energy ratios:** Porosity creates high-frequency bursts.
- **Spectral entropy:** Defects create acoustic disorder.
- **MFCC means, standard deviations, and delta/delta-delta:** Captures timbral characteristics and dynamics of the welding arc.
- **Temporal variance:** Arc instability means non-stationary sound.
- **Spectral contrast, spectral flatness, spectral bandwidth:** Capture differences in harmonic structure and noise-like components caused by defects.

Image/Video (128 features)

- **Laplacian variance:** Surface roughness of the bead.
- **Bead width consistency:** Defects cause irregular geometry.
- **Spatter detection:** Bright scattered spots outside weld zone.
- **GLCM texture:** Surface homogeneity vs. pitting.
- **Temporal frame differencing:** Motion energy during welding.
- **Edge orientation entropy:** Regular edges = good weld.

Total: **308 physics-informed features** per sample, each with physical justification.

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Chained XGBoost Pipeline

Our model uses a two-stage chained classifier for robust defect detection and classification.

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Stage 1: Binary Gate

Determines if a weld is defective or good using an XGBoost binary classifier with isotonic calibration.

- **Binary F1:** 0.902
- **AUC:** 0.934
- **Recall:** 0.940 (few missed defects)
- **Precision:** 0.867

Stage 2: Multiclass Assessor

Classifies the 6 defect types if a weld is deemed defective, trained only on defect samples.

- **Type Macro F1:** 0.819

Combined Final Score: 0.869 (0.6 × Binary_F1 + 0.4 × Type_MacroF1)

Key decisions: GroupKFold for data integrity, two-pass feature selection (93 features), 60-iteration RandomizedSearchCV, and multiprocessing for efficiency. Training was parallelized using **ThreadPoolExecutor** to speed up model fitting and hyperparameter searches.

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Dashboard, Inference & Results

We developed an end-to-end production pipeline for seamless operation.

- **One-command inference:** Automatically processes test data, detects modalities, and outputs results.
- **Graceful fallback:** Automatically switches to audio-visual mode if sensor data is missing.
- **Live dashboard:** Next.js frontend + FastAPI backend for data overview, per-sample drill-down, model predictions, and error analysis.

Per-Class Performance (115 unseen test samples)

Defect Type	F1 Score
Overlap (06)	0.933
Burn through (02)	0.743
Excessive convexity (08)	0.921
Good weld (00)	0.860
Lack of fusion (07)	0.709
Crater cracks (11)	0.805
Excessive penetration (01)	0.763

All defect types now achieve F1 > 0.70 on the unseen test set; the model reliably distinguishes seven classes with strong per-class performance, supporting a 7-class multiclass deployment.

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Why This Approach Wins

Physics-Grounded Features

Not a black box. Every feature maps to a physical defect mechanism, ensuring engineer trust.

Leakage-Proof Evaluation

GroupKFold by configuration folder ensures real-world generalization, not data memorization.

Production-Ready

One-command training/inference, live dashboard, and graceful fallback make it deployable.